VU MEDICAL ZONE

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Fermentation Technology

BT 603 Midterm ppt

Lecture 1 to 37

Advances in Fermentation Technology

Overview of Fermentation Technology-1

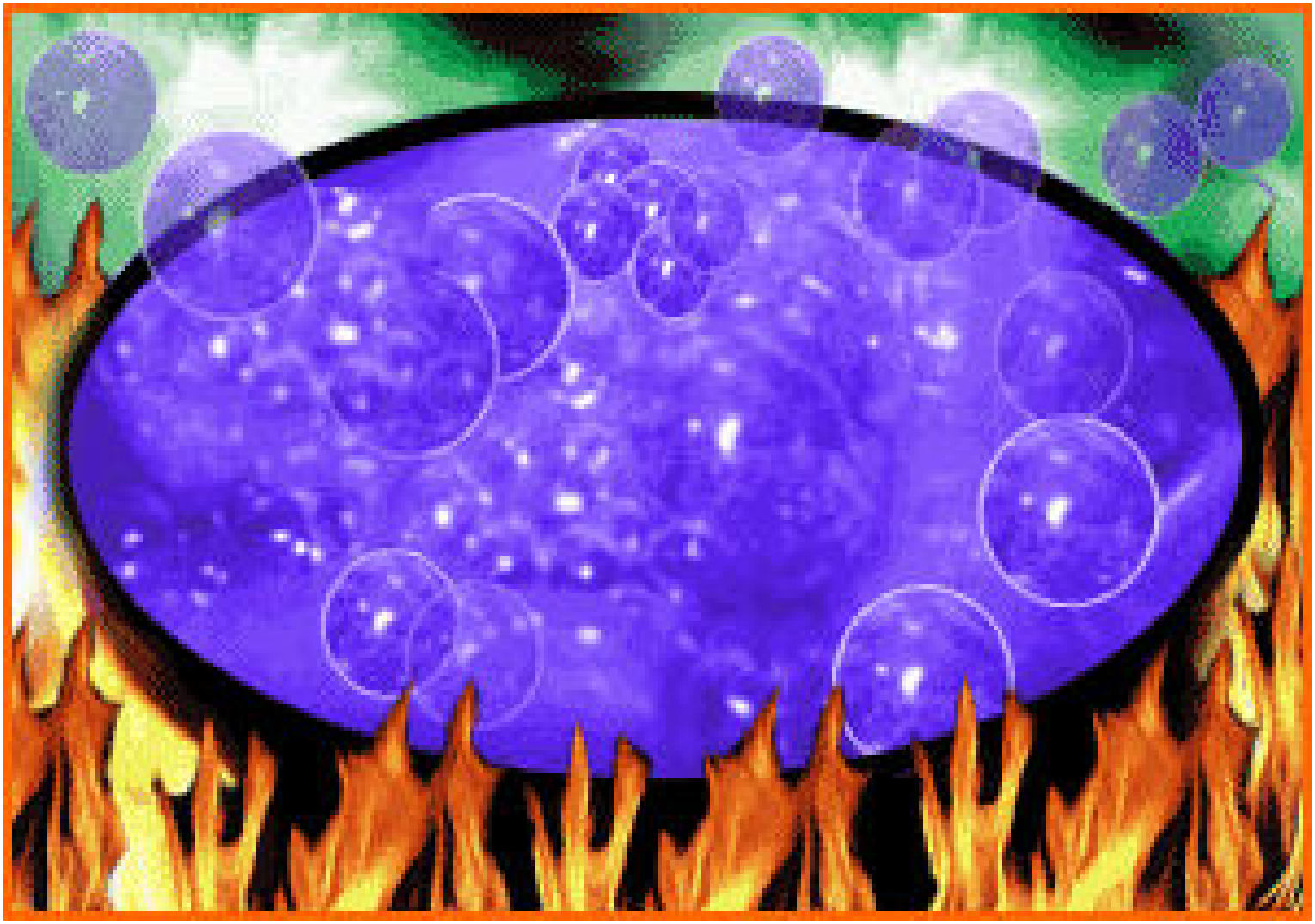
Overview of Fermentation Technology

WHAT IS FERMENTATION?

- Fermentation actually comes from Latin word "fervere" which means "to boil"

- So the word "Fermentation" is referred to physical state of boiling/bubbling.

Overview of Fermentation Technology



Overview of Fermentation Technology



Overview of Fermentation Technology

The bubbling appearance in fermentation
is due to the production of CO_2 bubbles caused by
anaerobic catabolism
of sugars present in the extract
by the action of yeast cells

Overview of Fermentation Technology

Definition of Fermentation

In the words of BIOCHEMIST

"Fermentation relates to the generation of energy by catabolism of organic compounds"

Overview of Fermentation Technology

But in strict biochemical sense

"the fermentation is Energy-generation process in which organic compounds act as both electron donors and electron acceptors"

Overview of Fermentation Technology

FERMENTATION IN TERMS OF A MICROBIOLOGIST:

INDUSTRIAL MICROBIOLOGISTS have extended the term fermentation much broader than CHEMISTS

Overview of Fermentation Technology

Microbiologists describe it as:

"any process for the production of product by the mass culture of micro-organism"

Overview of Fermentation Technology

FERMENTATION IN TERMS OF A BIOTECHNOLOGIST

The term fermentation becomes more broader than Mass culturing of the cells for the sake of desired product.

Overview of Fermentation Technology

Fermentation in biotechnology can be referred to the

"*Use of Biological system for the sake of development / improvement / facilitation in a process by mass culture of cells*"

END OF MODULE 1

Advances in Fermentation Technology

Overview of Fermentation Technology-2

Overview of Fermentation Technology

As the Fermentation is a process of mass culturing of the cells for the sake of desired product.

Overview of Fermentation Technology

By this definition, fermentation involves

1. *Mass culturing techniques*
2. *Types of cells and organisms*
3. *Fermentation Products*

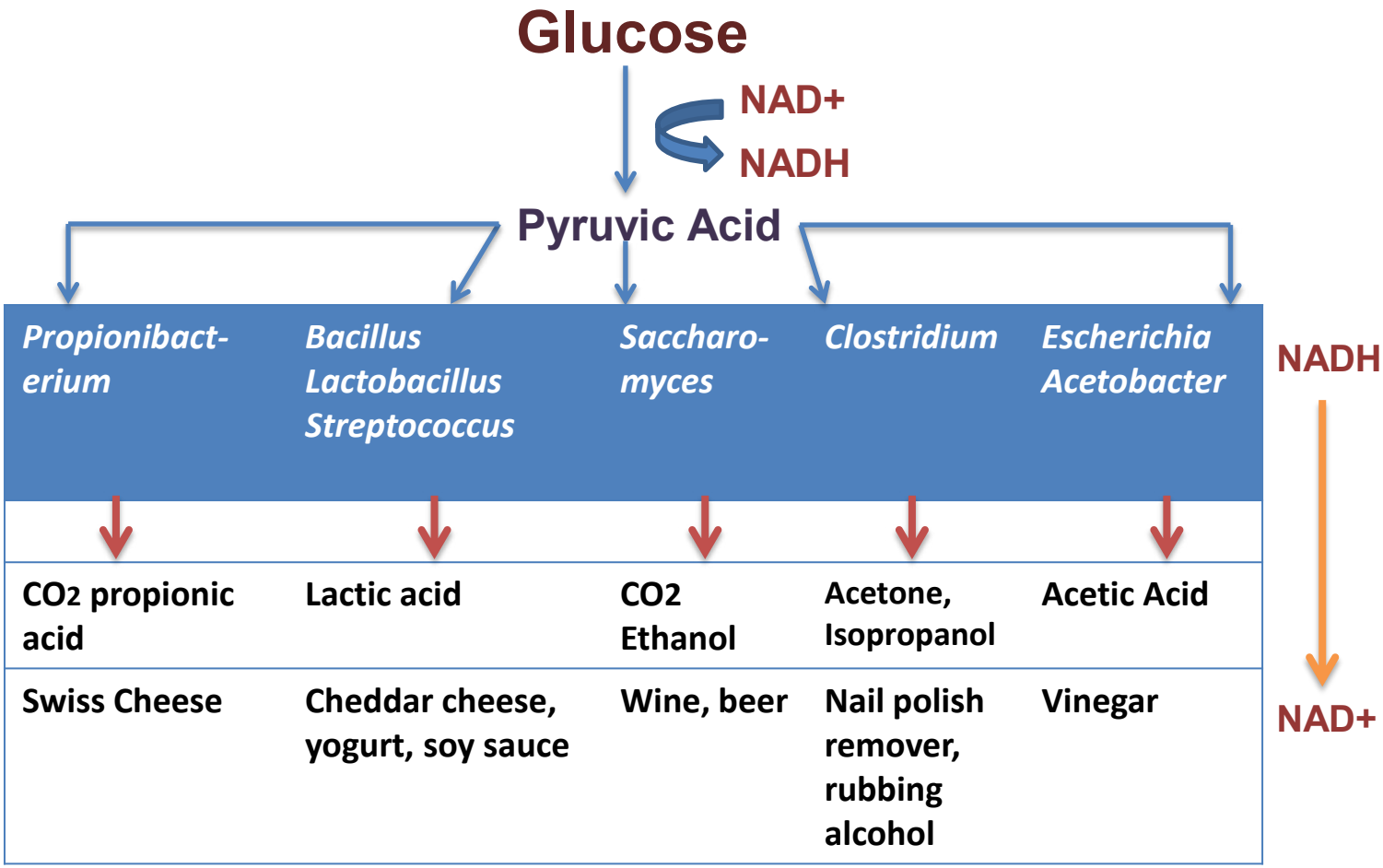
Overview of Fermentation Technology

Products of Fermentation Process

- What possible products we can made by fermentation process?

- What is the range of capacity of fermentation products?

Overview of Fermentation Technology



Overview of Fermentation Technology

Common Fermentation Products

Acids, Enzymes, Gasses, Alcohols, Hormones, Single Cell Protein, Yogurt, Cheese, Wines, Bread & Pickles, etc.

END OF MODULE 2

Advances in Fermentation Technology

Types of Fermentation

Types of Fermentation

Fermentation can be classified into different types on the basis of followings:

1. *Oxygen (O_2) Need*
2. *Moisture/Water (H_2O) content in medium*
3. *Production Scale*
4. *Process*
5. *State/Mixing*
6. *Product*
7. *Organism*

END

Advances in Fermentation Technology

Types of Fermentation on the basis of Oxygen (O_2) Need

Types of Fermentation

The O_2 supply must be according to need of the process of fermentation:

- **Aerobic Fermentation**
- **Semi-anaerobic Fermentation**
- **Anaerobic Fermentation**

Types of Fermentation

❑Aerobic Fermentation

- Aerobic fermentation is carried out in the presence of oxygen.
- Organisms use oxygen for the conversion of complex organic compounds.

Types of Fermentation

- **Aerobic Fermentation**

Oxygen here is actually required for the reproduction and growth of microorganisms.

Types of Fermentation

❑Semi-anaerobic Fermentation

Process in which organism makes ATP by aerobic respiration , but capable of switching to anaerobic respiration if oxygen is absent.

Types of Fermentation

- Semi-anaerobic Fermentation

Oxygen is needed, but not much more.

Types of Fermentation

❑Anaerobic Fermentation

Anaerobic fermentation is carried out in the **strict absence of oxygen**; means presence of oxygen may be toxic.

Types of Fermentation

- **Anaerobic Fermentation**

Involves Conversion of complex organic compounds into simpler ones in the absence of Oxygen.

END

Advances in Fermentation Technology

Types of Fermentation on the basis of Moisture/Water (H_2O) content in medium

Types of Fermentation

On the basis of water/ moisture requirement , process can be:

- **Solid State**
- **Submerged/ Liquid**
- **Surface Culture/ Jelly**

Types of Fermentation

❑Solid-State Fermentation (SSF)

In solid-state fermentation, the **microorganisms grow on a moist solid with little or no 'free' water**, although capillary water may be present.

Examples of this type of fermentation are seen in mushroom cultivation, bread-making and the processing of cocca, and in the manufacture of some traditional foods, e.g. miso (soy paste), soy sauce, tempeh (soybean cake) and gari (cassava), which are now produced in large industrial operations. SSF is also called as **Koji Fermentation.**

Types of Fermentation



Types of Fermentation

Some food fermentations involving SSF:

i. Wheat by *Aspergillus spp.*

ii. Soybean by *Rhizopus spp.*

iii. Soybean by *Aspergillus spp.*



Types of Fermentation

- **Submerged/Liquid Fermentation (SmF)**

Fermentation process wherein microorganisms employed grows in submerged state within the fermentation media.

Types of Fermentation

❑Submerged/Liquid Fermentation (SmF)

Submerged fermentation may **use a dissolved substrate**, e.g. sugar solution, **or a solid substrate, suspended in a large amount of water to form a slurry**.

Types of Fermentation



❑Submerged/Liquid Fermentation (SmF)

➢Organism is grown in a liquid medium which is vigorously aerated & agitated.

➢**Used** for pickling vegetables, brewing beer and producing wine and soy sauce.

Types of Fermentation

❑Surface Culture/ Jelly Fermentation

It refers to those fermentations in which the microorganisms utilized to grow on the surface of the fermentation media.

Types of Fermentation

❑Surface Culture/ Jelly Fermentation

➢Organism is allowed to grow on the surface of a liquid medium without agitation. After an appropriate incubation period the culture filtrate is separated from the cell mass and is processed to recover the desirable product.

➢**Citric acid** is commonly produced by surface fermentation process wherein the microorganism *Aspergillus niger* grows as a thick floating mycelial mat over the surface of the media used.

Types of Fermentation



Surface Culture Fermentation

END

Types
of
Fermentation
on the basis of
Production Scale

Types of Fermentation

On the basis of scale, there are following types of fermentation:

- **Laboratory scale fermentation**
- **Pilot scale fermentation**
- **Industrial scale fermentation**

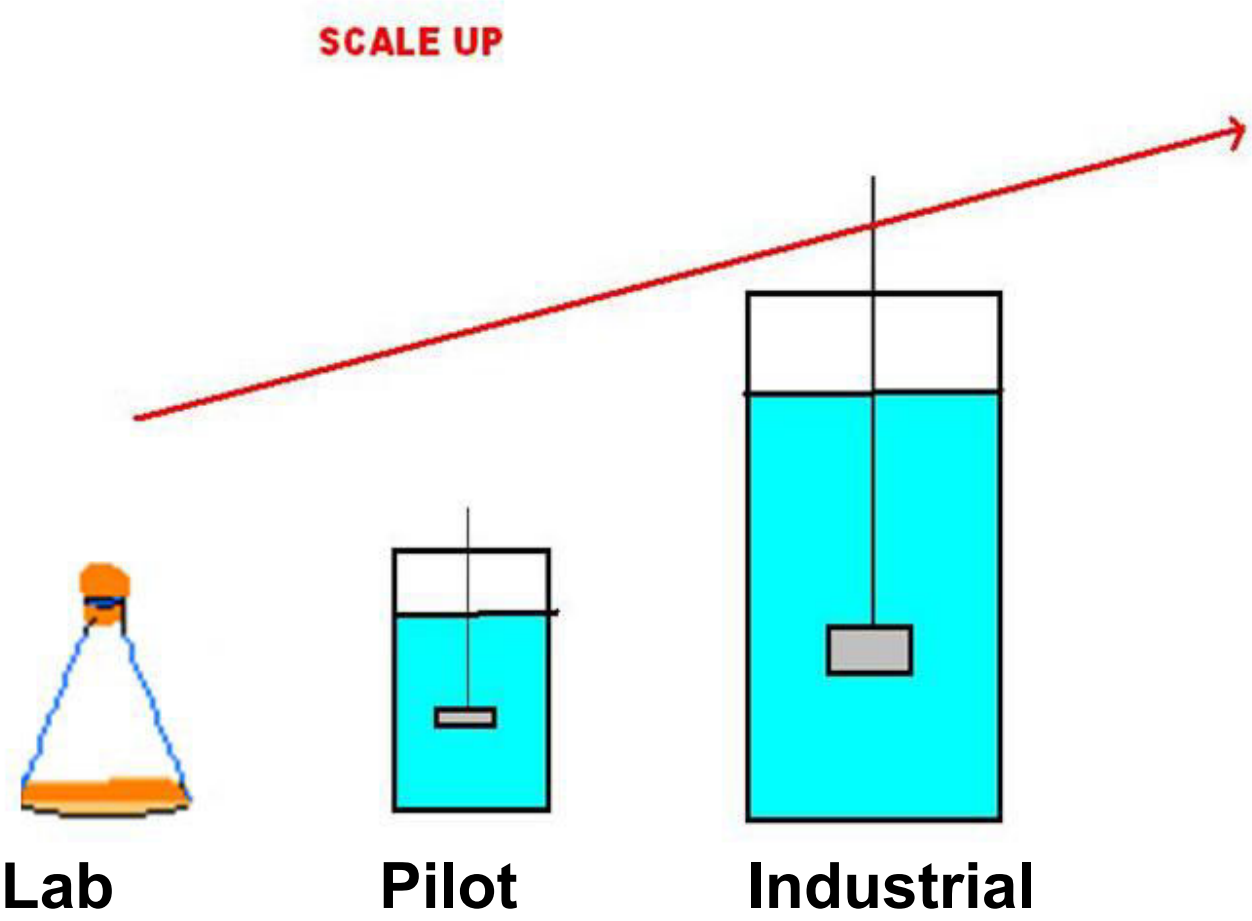
Types of Fermentation

- **Laboratory scale fermentation** is carried out in flasks, and cultures may be mixed by means of shaking in a controlled environment chamber.

- **Pilot and industrial scale fermentations** are normally carried out in stirred, aerated vessels, termed as fermenters.

Scale-up means increasing the production scale, for example from the laboratory scale to the pilot plant scale or from the pilot plant scale to industrial scale.

Types of Fermentation



Types of Fermentation

This is actually done on the basis of two factors:

- **Mode of sterilization**

In-situ → means sterilization is done on the spot/ inside fermenter maintaining high temperature by steam

Ex-situ → means sterilization is done outside e.g. in autoclave.

- **Inoculum Level/quantity**

An increase in scale also vary in the inoculum development .

Types of Fermentation

Ex-situ Sterilization in Autoclave





Types of Fermentation

In-situ Sterilization (Inside the Fermenter)



Types of Fermentation

- **Lab Scale Fermentation**

This is small scale fermentation mostly run at lab scale either in flask of different quantity or fermenter.

Types of Fermentation

❑Lab Scale Fermentation

- Sterilization mode at this scale is always ex-situ.
- Inoculum that runs fermentation process is prepared in slants/small flasks.

Types of Fermentation







Types of Fermentation

❑Pilot Scale Fermentation

A pilot scale plant offers flexibility, versatility and a broad range of equipment to rapidly convert laboratory results into industrial processes.

Types of Fermentation

- **Pilot Scale Fermentation**

In fact **pilot plant fermentation 'mimic' the industrial scale fermentation process** by carrying out the industrial process on a smaller scale.

Types of Fermentation

- We carry out fermentation plant studies based on **'uncertainties'** of what will the industrial scale fermentation plant will be like in reality.

AND

- How will the processes be similar or different from the smaller lab scale studies and how much changes will need to be executed at the various processes.

Types of Fermentation

Type of fermentations at inoculum level and mode of sterilization:

Inoculum			Sterilization	
Fermentation type	Slant/flask	Seed Fermenter	Ex-situ	In-situ
Lab Scale	✓		✓	
Pilot Scale	✓			✓
Industrial		✓		✓

Types of Fermentation

- Industrial Fermenters



END

Advances in Fermentation Technology

Range of Fermentation Products

Range of Fermentation Products

There are five major groups of commercially important fermentations:

1. Those that produce microbial cells (or biomass) as the product.
2. Those that produce microbial metabolites.
3. Those that produce microbial enzymes.
4. Those that produce recombinant products.
5. Those that modify a compound which is added to the fermentation – the transformation process.

Range of Fermentation Products

- **Biomass Production**
  - Baker's Yeast
  - SCP (Single Cell Protein)
  - Probiotics
  - Bio fertilizers
  - Vaccines
  - Bioweapons

Advances in Fermentation Technology

Range of Fermentation Products-2

Range of Fermentation Products

- **Biofertilizers**

Biofertilizers are low cost, renewable sources of plant nutrients which supplement chemical fertilizers.

Range of Fermentation Products

- **Biofertilizers**

These are nothing but selected strains of beneficial soil microorganisms cultured in the laboratory and packed in a suitable carrier. They can be used either for seed treatment or soil application.

Range of Fermentation Products

- **Bioweapons**
  - **Biological weapons** (often termed "bio-weapons", "biological threat agents", or "bio-agents") are living organisms or replicating entities (viruses, which are not universally considered "alive") that reproduce or replicate within their host victims.
  - The **North Atlantic Treaty Organization (NATO)** defines a biological weapons as "the provision of any infectious agent or toxin by any means of delivery in order to cause harm to humans, animals, or plants."

Range of Fermentation Products



❑**Top 5 Examples of Bioweapons**

1. Smallpox
2. Anthrax
3. Botulinum toxin
4. Aflatoxin
5. *Clostridium perfringens*

Range of Fermentation Products

❑Smallpox

If you took a gram of smallpox, which is highly contagious and lethal, and for which there's no vaccine available globally now, and released it in the air and created about a hundred cases, the chances are excellent that the virus would go global in six weeks as people moved from city to city.....the death toll could easily hit the hundreds of millions.....in scale, that's like a nuclear war.



Range of Fermentation Products

- **Anthrax**

Another Biological Weapon, ***B. anthracis***, is an aerobic spore forming, gram positive bacterium that is highly infections and lethal to man and many of his domestic animals.

Range of Fermentation Products

- **Anthrax**

Inhaled spores germinate in the lungs and produce a pulmonary anthrax which is rapidly fatal in 80% of the cases.



END

Range of Fermentation Products-3

Range of Fermentation Products

❑Metabolites Production

- Intermediates and products of metabolism.
- All the compounds and substances that are stops of metabolic reaction are metabolites.

Range of Fermentation Products

- **Metabolites Production**

Metabolites are classified as:

i. **Primary** Metabolites
ii. **Secondary** Metabolites
iii. **Tertiary** Metabolites

Range of Fermentation Products

❑Types of Metabolites

- **Primary Metabolites** are those which are directly involved in the process.

- **Secondary Metabolites** are those which are not directly involved in those metabolic processes, but usually have an important ecological function.

Metabolites may involve Acids e.g. Citric acid, Oxalic acid, Glutamic acid and Amino acid.

Range of Fermentation Products



❑Aflatoxins

This is a class of biological carcinogens, product by certain molds, that induce liver cancer. Molds that produces this material grows well on grain, peanuts and other rich nutrients.

Range of Fermentation Products

END

❑Botulinum Toxin

- Often touted as the most toxic substance in the world or at least in the biological world.
- Produced from *Clostridium botulinum*, an obligate anaerobe, which makes it a bit difficult to grow.

Advances in Fermentation Technology

Range of Fermentation Products-4

Range of Fermentation Products

❑Enzymes

- Enzymes are Functional Proteins, with few exceptions.
- Enzymes may be extracellular or intracellular.

Range of Fermentation Products

- **Enzymes**

Intracellular enzymes are those which remain within the cell and are responsible of catalyzing all those processes need in cells.

Range of Fermentation Products

- **Enzymes**

Extracellular enzymes are those which are released by the cells to perform a job external to the cell e.g. digestive enzymes.

Range of Fermentation Products

❑Enzymes

Enzymes can also be classified as constitutive enzymes or inducible enzymes.

- **Constitutive enzyme:** An enzyme that is synthesized continually regardless of growth conditions.
- **Inducible enzyme:** An enzyme that is synthesized only in the presence of the substrate that acts as an inducer.

Range of Fermentation Products

As example of inducible enzyme serves pyruvate carboxylase (enzyme which turns pyruvate into oxaloacetate) which is synthesized on demand and the process is activated by cortisol.

END

Advances in Fermentation Technology

Range of Fermentation Products-5

Range of Fermentation Products

❑Biotransformation

The conversion of molecules from one form to another within an organism.

Examples:

Starch → Maltose

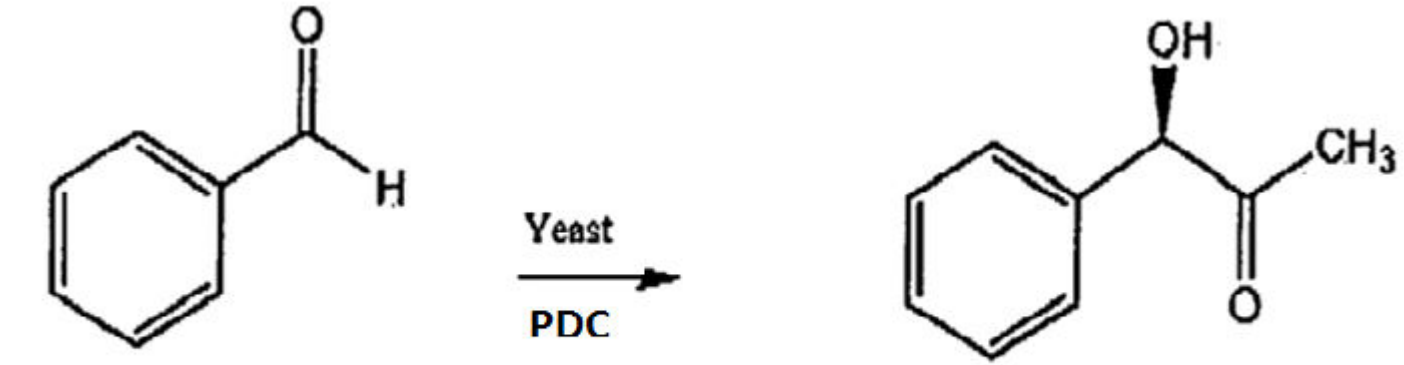
Starch → Glucose

Benzaldehyde → LPAC

Range of Fermentation Products

Biotransformation of Benzaldehyde to LPAC

Biotransformation of benzaldehyde to L-phenylacetylcarbinol (L-PAC) as a key intermediate for L-ephedrine synthesis has been synthesized by the bioconversion by yeast cells.



Benzaldehyde

L-Phenylacetylcarbinol (L-PAC)

Range of Fermentation Products

❑Biotransformation

For most microbial fermentations, glucose dissimilation occurs through the glycolytic pathway. A Simple organic compound most commonly generated is pyruvate, or a compound derived enzymatically from pyruvate, such as acetaldehyde, etc.

Range of Fermentation Products

- **Biotransformation**

Acetaldehyde can then be reduced by NADH + H^+ to ethanol, which is excreted by the cell.

END

Range of Fermentation Products-6

Range of Fermentation Products

❑Recombinant Products

The advent of rDNA technology has extended the range of fermentation products. Genes from higher organisms may be introduced into microbial cells with suitable expression systems.

Range of Fermentation Products

❑Recombinant Products

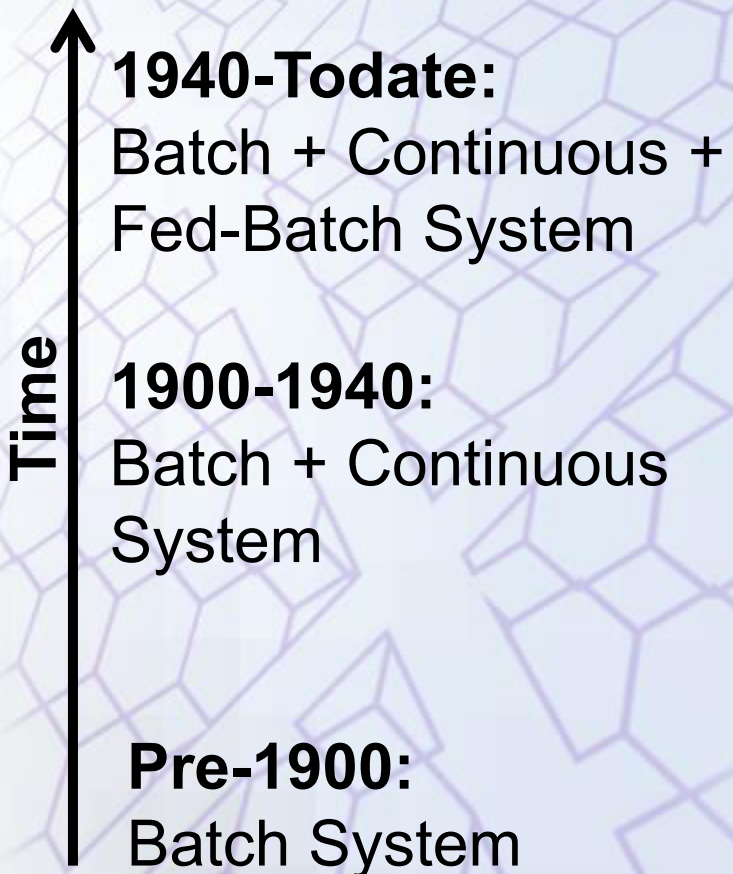
- Interferons
- Insulin
- Human Serum Albumin
- Clotting Factors VIII and IX
- Epidermal Growth Factor
- Calf Chymosin
- Bovine Somatostatin

Range of Fermentation Products

Stages in the chronological development of fermentation industry

Stage	Main Products
(1) Pre-1900	Alcohol and Vinegar.
(2) 1900-1940	Baker's Yeast, Glycerol, Citric acid, Lactic acid, Acetone/Butanol.
(3) 1940-date	Penicillin, Streptomycin, Gibberelin, aminoacids, nucleotides, enzymes.
(4) 1964-date	Single-cell Protein using hydrocarbons and other feed stocks.
(5) 1979-date	Production of heterologous proteins by microbial and animal cells, Mabs production by animal cells.

Range of Fermentation Products



END

Advances in Fermentation Technology

Types of Fermentation on the basis of

Process

Types of Fermentation

- On the basis of process, fermentation can be

1) Batch fermentation
2) Fed-Batch Fermentation
3) Continuous Fermentation

Types of Fermentation

❑Batch Fermentation

- It is a closed system.
- In batch processing, a batch of culture medium in a fermenter is inoculated with a microorganism (the 'starter culture'). The fermentation proceeds for a certain duration (the 'fermentation time' or 'batch time'), and the product is harvested at the end.
- In the course of the entire fermentation, nothing is added except oxygen (in the form of air, if the process is aerobic), an antifoam agent, and acid or base to control the pH. The culture is allowed to grow until no more of the product is being made, at which point the reactor is "harvested" and cleaned out for another run.

Types of Fermentation

❑Fed-Batch Fermentation

➢An enhancement of the closed batch process is the fed-batch fermentation.

➢In fed-batch fermentations, sterile culture medium is added either continuously or periodically to the inoculated fermentation batch.

➢Nutrients in the medium are added when needed and can't be added in bulk amount at start because they can act as toxin in bulk amounts.

➢The volume of fermented broth increases with each addition of the medium, and the fermenter is harvested after the batch time.

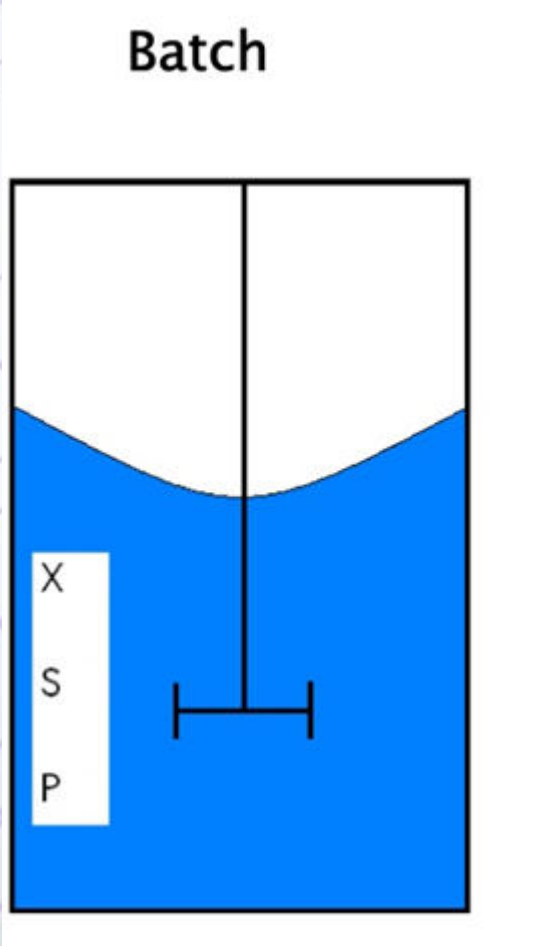
Types of Fermentation

❑Continuous Fermentation

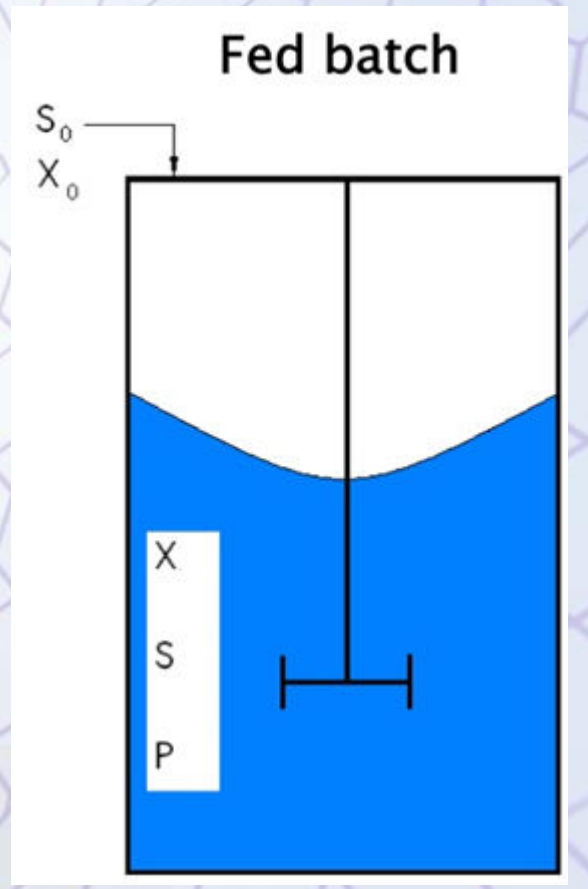
➢In continuous fermentations, sterile medium is fed continuously into a fermenter and the fermented product is continuously withdrawn, so the **fermentation volume remains unchanged.**

➢Typically, continuous fermentations are started as batch cultures and feeding begins after the microbial population has reached a certain concentration.

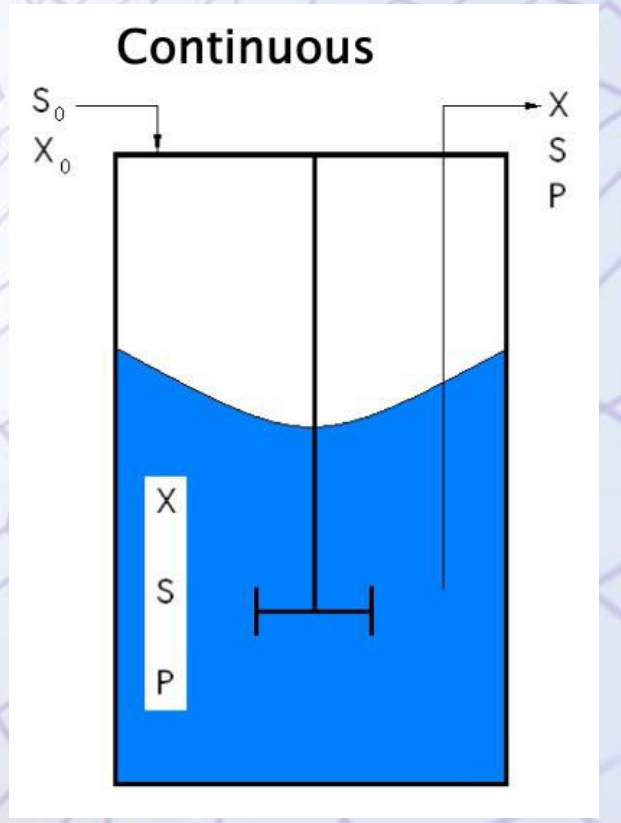
Types of Fermentation



Types of Fermentation



Types of Fermentation



END

Types of Fermentation on the basis of State of Mixing

Types of Fermentation

➢ **Mixing in Fermentation is done:**

1) To disperse the air bubbles.
2) To suspend the cells.
3) To enhance heat and mass transfer in the medium.

Types of Fermentation

- **On the basis of state or mixing fermentation can be**

1) Static Fermentation
2) Shake-Flask Fermentation
3) Stirred Fermentation

Types of Fermentation

❑Static Fermentation

➢In this type of fermentation there is no mixing during the process.

➢Surface culture

➢Un-aerated and unmixed bioreactor can be used here.

Types of Fermentation

- **Static Fermentation**

Un-aerated and unmixed bioreactor is the one in which the bed is static, or mixed only very occasionally (i.e., once or twice per day), and air is circulated around the bed but not blown forcefully.

Types of Fermentation

❑Shake-Flask Fermentation

On small level, simple shaking procedure can be applied during fermentation process, hence termed as shake flask fermentation. Shake culture flasks are usually applied to aerobic processes.

Types of Fermentation

❑Shake-Flask Fermentation

Shake cultures are a convenient method of growing microorganisms in submerged cultures under aerobic conditions created by shaking; it is a small scale equivalent of stirred tank bioreactor.

Types of Fermentation

Shake-Flask Fermentation



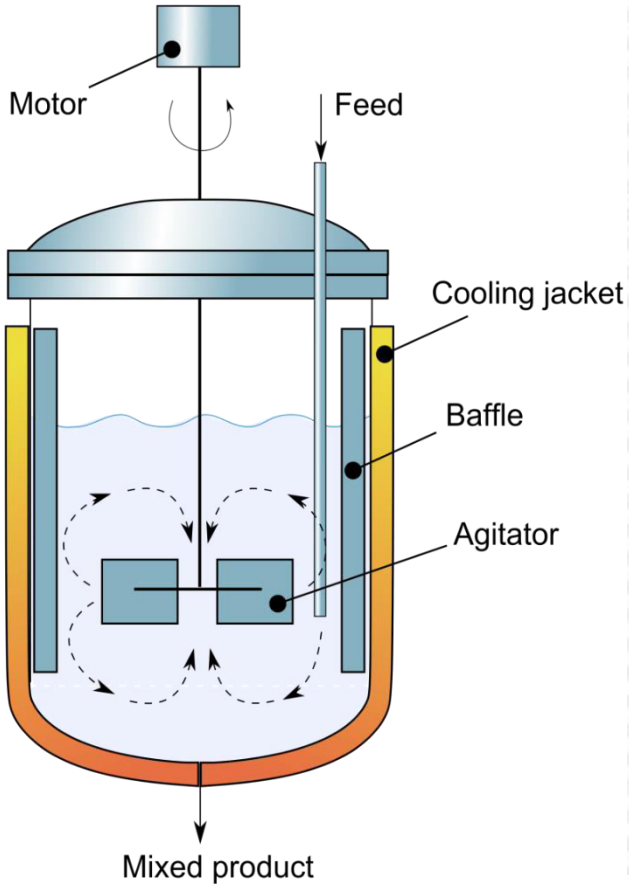
Types of Fermentation

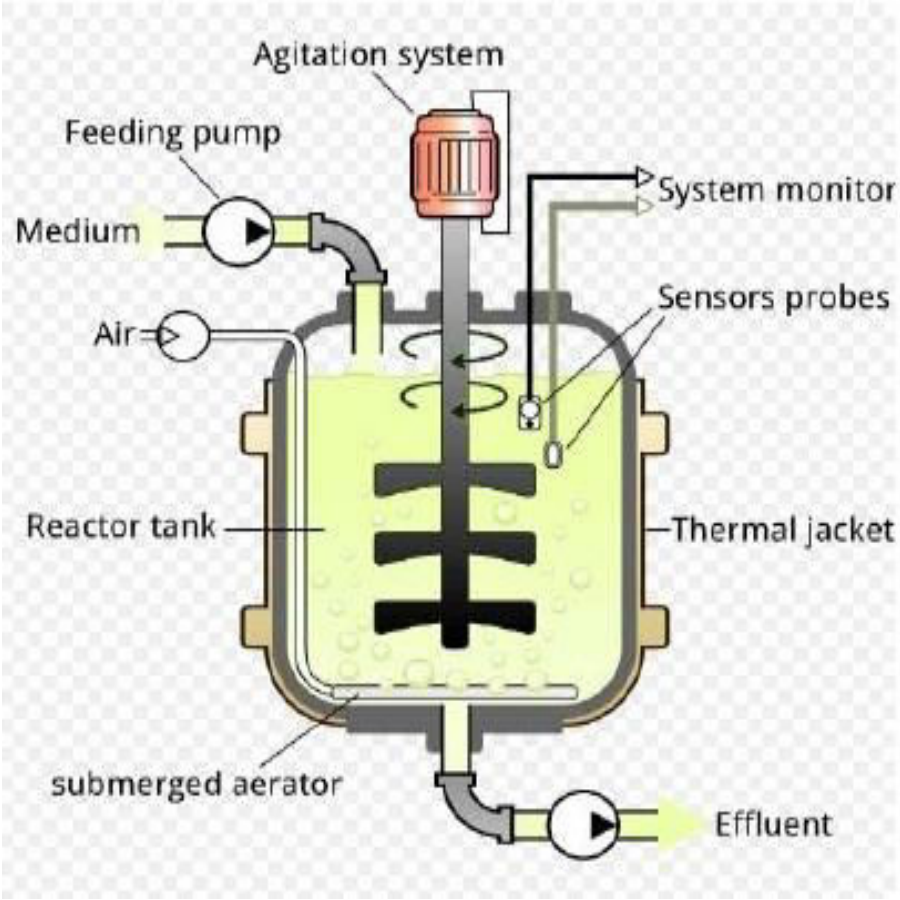
- **Stirred Fermentation**

This type of fermentation involves proper mixing during the process of fermentation which is done by: Agitator/ Impeller.

Types of Fermentation

Stirred Fermenters





Types of Fermentation

- **Stirred Fermentation**

The medium must be stirred to keep the cells in suspension Various types of stirrers range from **simple magnetic stirrers, flat blade turbine impellers, to marine impellers.**

END

Advances in Fermentation Technology

Types of Fermentation on the basis of

Product & Organisms

Types of Fermentation

Following are the types of fermentation on the basis of the products that formed:

1) Beer Fermentation
2) Yogurt Fermentation
3) Citric acid Fermentation
4) Pickle Fermentation

and many more

Types of Fermentation

Fermentation can also be named on the basis on the organisms nomenclature. For example:

1) Yeast Fermentation
2) Bacterial Fermentation
3) *Aspergillus niger* Fermentation

and many more

END

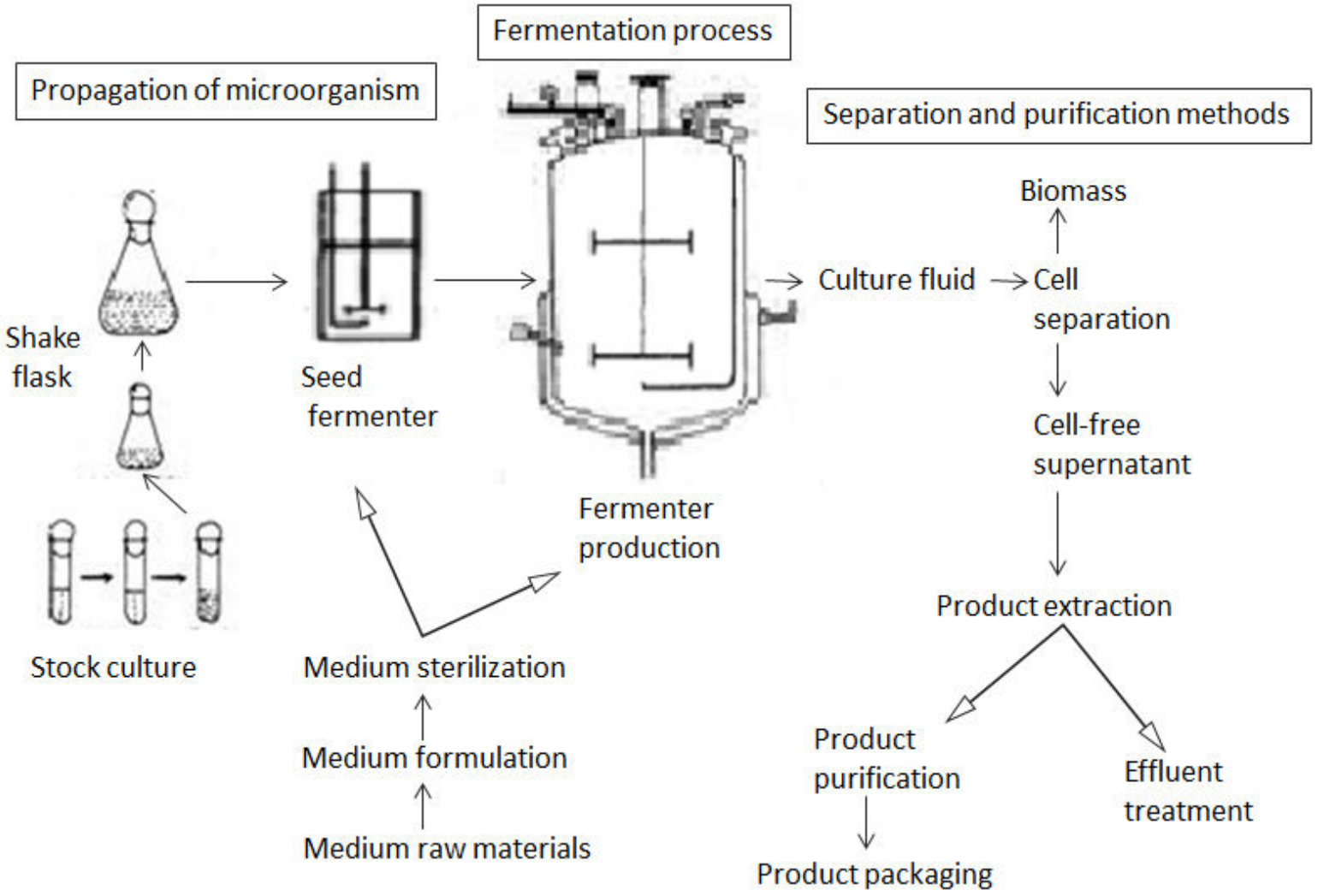
Organization of Fermentation Process

Organization of Fermentation Process

Regardless of the type of fermentation, an established process may be divided into following basic component parts:

1. **Organism** (Hero of Fermentation process).
2. **Fermenter** (stage on which microorganism act/grow).
3. **Media** (Formulation).
4. **Sterilization.**
5. **Monitoring/Control** (Environmental conditions).
6. **Recovery of the product** (Isolation, purification etc).
7. **Effluent treatment** (processing of materials that remained after fermentation process.

Organization of Fermentation Process



Organization of Fermentation Process

Fermentation process has two sub-processes that are:

1) ***Up-streaming Process***
   (Step 1-5)
2) ***Down-streaming Process***
   (Step 6-7)

Organization of Fermentation Process

❑Up-streaming Process

Upstream processing is an industrial term describing the **steps followed for the production of a specific product** by industrial means.

Organization of Fermentation Process

❑Down-streaming Process

Downstream Processing includes the **recovery of the products** in a pure state and the **effluent treatment**.

Organization of Fermentation Process

❑Down-streaming Process

Product recovery is carried out through a series of operations including ***cell separation*** by settling, centrifugation or filtration.

Organization of Fermentation Process

❑**Down-streaming Process**

Product recovery by disruption of cells (if the product intracellular); ***Extraction*** and ***purification*** of the product.

Organization of Fermentation Process

❑**Down-streaming Process**

Finally, the ***effluents are treated*** by chemical, physical or biological methods.

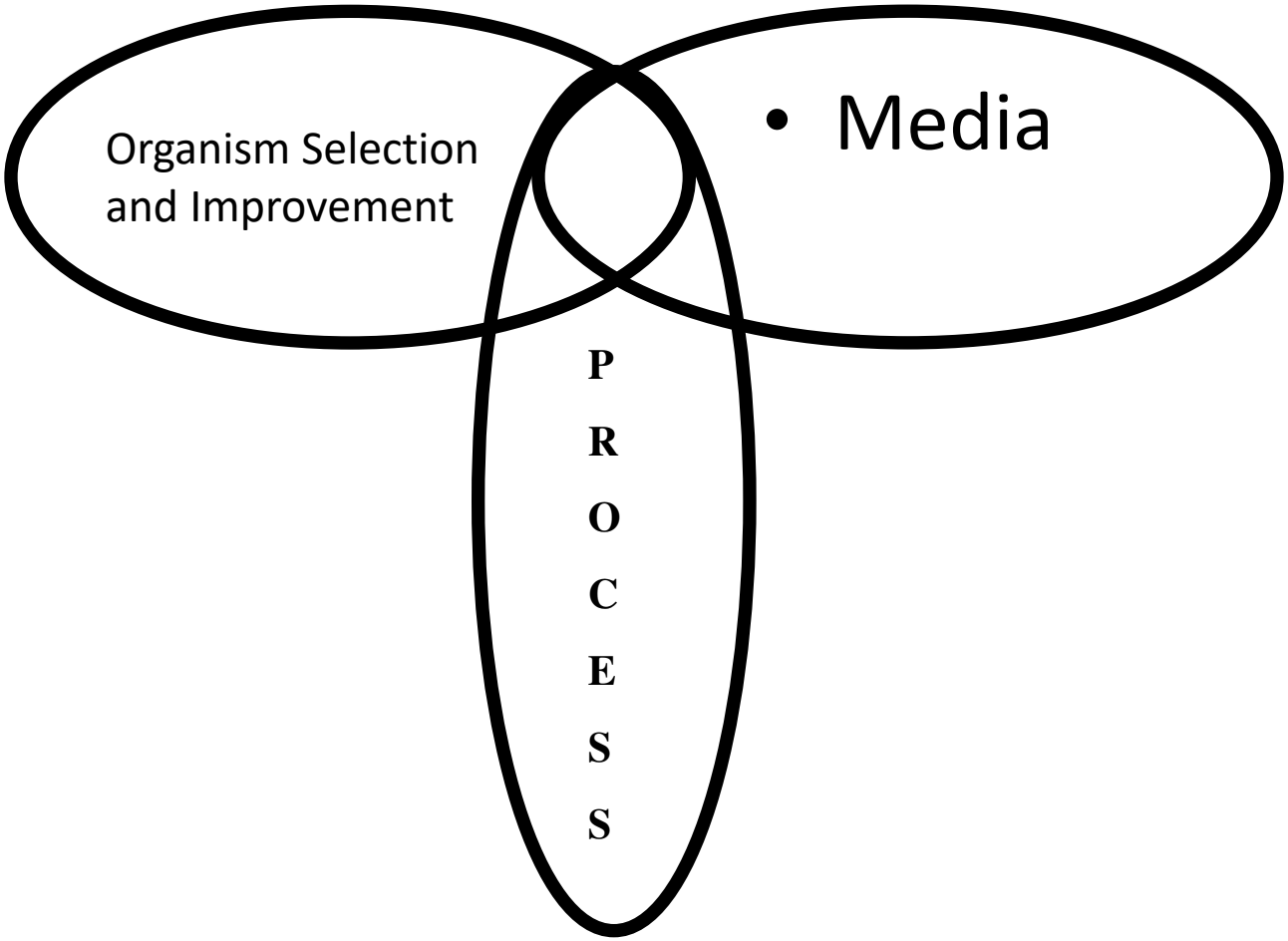
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Media for Industrial Fermentations:

Introduction-1

Media for Industrial Fermentations

Overview



Media for Industrial Fermentations

Medium

The intervening substances through which metabolic requirements are conveyed or physical forces are transmitted.

Media for Industrial Fermentations

- What does the medium need to do?

  Grow the microorganism so it produces biomass and product and should not interfere with down stream processing

Media for Industrial Fermentations

Detailed investigation is needed to establish the most suitable medium for an individual fermentation process, but certain basic requirements must be met by any such medium.

All micro-organisms require

- Water,
- Sources of energy,
- Carbon,
- Nitrogen,
- Mineral elements ,

and possibly

- Vitamins,

plus

- Oxygen (if aerobic) or other gasses such as CO_2 etc.

Media for Industrial Fermentations

It must be remembered that the medium will affect the design of fermenter to be used.

END

Media for Industrial Fermentations:

Introduction-2

Media for Industrial Fermentations

On a small scale it is relatively simple to devise a medium containing pure compounds, but the resulting medium, although supporting satisfactory growth, may be unsuitable for use in a large scale process.

On a large scale one must normally use sources of nutrients to create a medium which will meet as many as possible of the following criteria:

Media for Industrial Fermentations

1. It will produce the maximum yield of product or biomass per gram of substrate used.
2. It will produce the maximum concentration of product or biomass.
3. It will permit the maximum rate of product formation.
4. There will be the minimum yield of undesired products.
5. It will be of a consistent quality and be readily available throughout the year.
6. It will cause minimal problems during media making and sterilization.
7. It will cause minimal problems in other aspects of the production process particularly aeration and agitation, extraction, purification and waste treatment.

Media for Industrial Fermentations

- Crude and defined media:
  - Crude media is made up of unrefined agricultural products e.g. containing barley.
  - Defined media are like those we use in the lab e.g. minimal salts medium.
  - Crude media is cheap but composition is variable.
  - Defined media is expensive but composition is known and should not vary.
  - Crude media is used for large volume inexpensive products e.g. biofuel from whey.
  - Defined media is used for expensive low volume products e.g. anticancer drugs.

Media for Industrial Fermentations

The use of cane molasses, beet molasses, cereal grains, starch, glucose, sucrose and lactose as carbon sources, and ammonium salts, urea, nitrates, corn steep liquor, soya bean meal, slaughter-house waste and fermentation residues as nitrogen sources, have tended to meet most of the above criteria for production media because they are cheap substrates.

Media for Industrial Fermentations

However, other more expensive pure substrates may be chosen if the overall cost of the complete process can be reduced because it is possible to use simpler procedures.

END

Media for Industrial Fermentations:

Introduction-3

Media for Industrial Fermentations

- The problem of developing a process from the laboratory to the pilot scale, and subsequently to the industrial scale, must also be considered for selecting a suitable medium.
- A laboratory medium may not be ideal in a large fermenter with a low gas-transfer pattern. A medium with a high viscosity will also need a higher power input for effective stirring. Besides meeting requirements for growth and product formation, the medium may also influence pH variation, foam formation, the oxidation-reduction potential, and the morphological form of the organism.
- It may also be necessary to provide precursors or metabolic inhibitors. The medium will also affect product recovery and effluent treatment.

Media for Industrial Fermentations

Historically, undefined complex natural materials have been used in fermentation processes because they are much cheaper than pure substrates. However, there is often considerable batch variation because of variable concentrations of the component parts and impurities in natural materials which cause unpredictable biomass and/or product yields.

As a consequence of these variations in composition small yield improvements are difficult to detect. Undefined media often make product recovery and effluent treatment more problematical because not all the components of a complex nutrient source will be consumed by the organism.

The residual components may interfere with recovery and contribute to the BOD (Biological Oxygen demand) of the effluent.

Media for Industrial Fermentations

Thus, although manufacturers have been reluctant to use defined media components because they are more expensive, pure substrates give more predictable yields from batch to batch and recovery.

Media for Industrial Fermentations

Purification and effluent treatment are much simpler and therefore cheaper.

Process improvements are also easier to detect when pure substrates are used.

END

Media for Industrial Fermentations: Introduction-4

(General Types of Media)

Media for Industrial Fermentations

Selective media are designed to suppress the growth of unwanted bacteria and encourage the growth of the desired microbes. For example, **bismuth sulfite** is one medium used to isolate the typhoid bacterium, the gram-negative *Salmonella typhi*. Bismuth sulfite inhibits gram-positive bacteria and most gram-negative intestinal bacteria (other than *S. typhi*), as well. **Sabouraud's dextrose**, which has a pH of 5.6, is used to isolate fungi that outgrow most bacteria at this pH. **Starch** is used for selective and differential isolation of amylase/starch degrading enzymes producing microorganisms.

Media for Industrial Fermentations

Differential media make it easier to distinguish colonies of the desired organism from other colonies growing on the same plate. Similarly, pure cultures of microorganisms have identifiable reactions with differential media in tubes or plates. **Blood medium** (which contains red blood cells) is a medium that microbiologists often use to identify bacterial species that destroy red blood cells. These species, such as *Streptococcus pyogenes*, the bacterium that causes strep throat, show a clear ring around their colonies (beta-hemolysis) where they have lysed the surrounding blood cells

Media for Industrial Fermentations

Enrichment culture is used when bacteria present in small numbers can be missed, especially if other bacteria are present in much larger numbers, it is sometimes necessary to use an enrichment culture. This is often the case for soil samples. The medium (enrichment medium) for an enrichment culture is usually liquid and provides nutrients and environmental conditions that favour the growth of a particular microbe but not others. In this sense, it is also a selective medium, but it is designed to increase very small numbers of the desired type of organism to detectable levels.

Media for Industrial Fermentations

The cultivation of anaerobic bacteria poses a special problem.
Because anaerobes might be killed by exposure to oxygen, special media called Reducing media must be used.

Media for Industrial Fermentations

Reducing media contain ingredients, such as sodium thioglycolate, that chemically combine with dissolved oxygen and deplete the oxygen in the culture medium.

END

Media for Industrial Fermentations:

Introduction-5

Media for Industrial Fermentations

Collins (1990) has given an excellent example of a process producing recombinant protein from S. *cerevisiae* instead of just biomass. The range of growth conditions which can be used is restricted because of factors affecting the stability of the recombinant protein.

The control of pH and foam during growth in a fermenter were indentified as two important parameters. Molasses would normally be used as the cheapest carbohydrate to grow yeast biomass in a large scale process. However, this is not acceptable for the recombinant protein production because of the difficulties, and incurred costs caused in subsequent purification which result from using crude undefined media components.

Media for Industrial Fermentations

Collins and co-workers therefore used a defined medium with glucose, sucrose or another suitable carbon source of reasonable purity plus minimal salts, trace elements, pure vitamins and ammonia as the main nitrogen source and for pH control. Other impurities in molasses might have helped to stabilize foams and led to the need to use antifoams.

Tables in next slides gives the recipes for some typical media for submerged culture fermentations.

Media for Industrial Fermentations

Itaconic acid (Nubel and Ratakak, 1962)

Cane Molasses (as sugar)	150 gdm^{-3}
$ZnSO_4$	1.0 gdm^{-3}
$ZnSO_4.7H_2O$	3.0 gdm^{-3}
$CuSO_4.5H_2O$	0.01 gdm^{-3}

Amylase (Underkofler, 1966)

Ground soybean meal	1.85%
Autolysed Brewers yeast fractions	1.50%
Distilled dried solubles	0.76%
Enzymatic casein hydrolysate	0.65%
Lactose	4.75%
$MgSO_4.7H_2O$	0.04%
Hodag KG-1 antifoam	0.05%

Media for Industrial Fermentations

Glutamic acid (Gore *et al.*, 1968)

Dextrose	270 gdm^{-3}
$NH_4H_2PO_4$	2.0 gdm^{-3}
$(NH_4)_2HPO_4$	2.0 gdm^{-3}
K_2SO_4	2.0 gdm^{-3}
$MgSO_4.7H_2O$	0.5 gdm^{-3}
$MnSO_4.H_2O$	0.04 gdm^{-3}
$FeSO_4.7H_2O$	0.02 gdm^{-3}
Polyglycol 2000	0.3 gdm^{-3}
Biotin	12 μgdm^{-3}
Penicillin	11 μgdm^{-3}

Media for Industrial Fermentations

These examples of different media (in previous slides) are used to illustrate the range of media in use, but are not necessarily the best media in current use.

END

Media for Industrial Fermentations:

Medium Formulation-1

Media for Industrial Fermentations

Medium Can Be a Significant Proportion of Total Product Cost

Elements of total product cost (%)

Item	Product					
	Beer (Pratten, 1971)	Alcohol (batch) (Maiorella, 1984)	Acetic acid (Pape, 1977)	Citric acid (Schierholt, 1977)	SCP (Anon, 1974)	Penicillin (Swartz 1979)
Raw materials	38.4	76.7	42.2	39.7	62	58.0
Utilities	*	11.7	23.1	35.3	10	20.3
Labour and supervision	24.5	2.9	19.5	25.0	9	5.4
Fixed charges	7.2	4.8	10.5	—	19	—
Maintenance	29.9	—	—	—	†	14.9
Operating supplies	—	—	—	—	—	1.4‡

Raw materials costs range from 38-77% in the examples shown

Media for Industrial Fermentations

Medium formulation is an essential stage in the design of successful laboratory experiments, pilot-scale development and manufacturing processes. The constituents of a medium must satisfy the elemental requirements for cell biomass and metabolite production and there must be an adequate supply of energy for biosynthesis and cell maintenance. The first step to consider is an equation based on the stoichiometry for growth and product formation. Thus for an anaerobic fermentation:

C + N+ O_2 +other req -- biomass + product + CO_2 + H_2O + Heat

Media for Industrial Fermentations

The above equation should be expressed in quantitative terms which is important in the economical design of media component wastage is to be minimal.

Thus, it should be possible to calculate the minimal quantities of nutrients which will be needed to produce a specific amount of biomass.

Knowing that a certain amount of biomass is necessary to produce a defined amount of product, it should be possible to calculate substrate concentrations necessary to produce required product yields.

There may be medium components which are needed for product formation which are not required for biomass production. Unfortunately, it is not always easy to quantify all the factors very precisely.

Media for Industrial Fermentations

A knowledge of the elemental composition of a process microorganism is required for the solution of the elemental balance equation. Table given in next slide will serve as a guide to the absolute minimum quantities of N, S, P, Mg and K to include in an initial medium recipe.

Trace elements (Fe, Zn, Cu, Mn, Co, Mo, B) may also be needed in smaller quantities.

The concentration of P is deliberately raised in many media to increase the buffering capacity. These points emphasize the need for considerable attention to be given to medium design.

Media for Industrial Fermentations

TABLE *Element composition of bacteria, yeasts and fungi (% by dry weight)*

Element	Bacteria (Luria, 1960; Herbert, 1976; Aiba *et al.*, 1973	Yeasts (Aiba *et al.*, 1973; Herbert, 1976)	Fungi (Lilly, 1965; Aiba *et al.*, 1973)
Carbon	50–53	45–50	40–63
Hydrogen	7	7	
Nitrogen	12–15	7.5–11	7–10
Phosphorus	2.0–3.0	0.8–2.6	0.4–4.5
Sulphur	0.2–1.0	0.01–0.24	0.1–0.5
Potassium	1.0–4.5	1.0–4.0	0.2–2.5
Sodium	0.5–1.0	0.01–0.1	0.02–0.5
Calcium	0.01–1.1	0.1–0.3	0.1–1.4
Magnesium	0.1–0.5	0.1–0.5	0.1–0.5
Chloride	0.5	—	—
Iron	0.02–0.2	0.01–0.5	0.1–0.2

Media for Industrial Fermentations

Some micro-organisms cannot synthesize specific nutrients, e.g. amino acids, vitamins or nucleotides. Once a specific growth factor has been identified it can be incorporated into a medium in adequate amounts as a pure compound or as a component of a complex mixture.

END

Media for Industrial Fermentations:

Medium Formulation-2

Media for Industrial Fermentations

- Typical medium ingredients:
  - Water
  - Carbon sources and energy sources
  - Nitrogen sources
  - Vitamins and growth factors
  - Minerals and trace elements
  - Buffers
  - Inducers
  - Precursors
  - Chelators
  - Inhibitors e.g. KMS in beer medium
  - Antifoams

Media for Industrial Fermentations

The carbon substrate has a dual role in biosynthesis and energy generation. The carbon requirement for biomass production under aerobic conditions may be estimated from the cellular yield coefficient *(Y)* which is defined as:

$$\text{cellular yield coefficient } (Y) = \frac{\text{Quantity of cell dry matter produced}}{\text{Quantity of carbon substrate utilized}}$$

An adequate supply of the carbon source is essential for a product-forming fermentation process.

Media for Industrial Fermentations

TABLE *Cellular yield coefficients (Y) of bacteria on different carbon substrates (data from* Abbott and Clamen, 1973)

Substrate	Cellular yield coefficient (g biomass dry wt. g^{-1} substrate)
Methane	0.62
n-Alkanes	1.03
Methanol	0.40
Ethanol	0.68
Acetate	0.34
Malate	0.36
Glucose (molasses)	0.51

Media for Industrial Fermentations

According to the values from Table in previous slide, for bacteria with a *Y* for glucose of 0.5, which is 0.5 g cells g^{-1} glucose, the conc. of glucose needed to obtain 30 g dm^{-3} cells will be 30/0.5=60 g dm^{-3} glucose. One litre of this medium would also need to contain approximately 3.0 g N, 1.0 g P, 1.0 g K, 0.3 g S and 0.1 g Mg.

END

Media for Industrial Fermentations:

Medium Formulation-3

Media for Industrial Fermentations

Cooney (1979) has calculated theoretical yields for Penicillin G biosynthesis on the basis of material and energy balances using a biosynthetic pathway based on reaction stoichiometry. The stoichiometry equation for the overall synthesis is:

$$a_2C_6H_{12}O_6 + b_2NH_3 + c_2O_2 + d_2H_2SO_4 + e_2PAA \rightarrow n_2\text{Pen G} + p_2CO_2 + q_2H_2O$$

where a_2, b_2, c_2, d_2, e_2, n_2, p_2 and q_2 are the stoichiometric coefficients and PAA is phenylacetic acid.

Media for Industrial Fermentations

Solution of above equation yields:

$$10/6\ C_6H_{12}O_6 + 2NH_3 + 1/2\ O_2 + H_2SO_4$$
$$+ C_8H_8O_2 \rightarrow C_{16}H_{18}O_4N_2S + 2CO_2 + 9H_2O$$

In this instance it was calculated that the theoretical yield was 1.1 g penicillin G g^{-1} glucose (1837 units mg^{-1}).

The other major nutrient which will be required is oxygen which is provided by aerating the culture.

Media for Industrial Fermentations

Medium design will influence the oxygen demand of a culture in that the more reduced carbon sources will result in a higher oxygen demand.

END

Media for Industrial Fermentations:

Medium Components

Media for Industrial Fermentations

Water is the major component of all fermentation media, and is needed in many of the ancillary services such as

- ✓ heating
- ✓ cooling
- ✓ cleaning
- ✓ rinsing.

When assessing the suitability of a water supply it is important to consider pH, dissolved salts and effluent contamination.

The mineral content of the water is very

Media for Industrial Fermentations

Energy for growth comes from either the oxidation of medium components or from light. Most industrial microorganisms are chemo-organotrophs, therefore the commonest source of energy will be the carbon source such as carbohydrates, lipids and proteins.

Media for Industrial Fermentations

Some micro-organisms can also use hydrocarbons or methanol as carbon and energy sources.

END

Media for Industrial Fermentations:

CARBON SOURCES

Media for Industrial Fermentations

❑Factors influencing the choice of carbon source

The rate at which carbon source is metabolized influence the formation of biomass or production of primary or secondary metabolites. Fast growth due to high concentrations of rapidly metabolized sugars is often associated with low productivity of secondary metabolites.

The main product of a fermentation process will often determine the choice of carbon source, particularly if the product results from the direct dissimilation of it.

In fermentations such as ethanol or single-cell protein production where raw materials are 60 to 77% of the production cost, the selling price of the product will be determined largely by the cost of the carbon source.

Media for Industrial Fermentations

❑Factors influencing the choice of carbon source

The method of media preparation, particularly sterilization, may affect the suitability of carbohydrates for individual fermentation processes. It is often best to sterilize sugars separately because they may react with ammonium ions and amino acids to form black nitrogen containing compounds which will partially inhibit the growth of many micro-organisms.

Starch suffers from the handicap that when heated in the sterilization process it gelatinizes, giving rise to very viscous liquids, so that only concentrations of up to 2% can be used without modification.

The choice of substrate may also be influenced by government legislation.

Media for Industrial Fermentations

The purity of the carbon source may also affect the choice of substrate. For example, metallic ions must be removed from carbohydrate sources used in some citric acid processes.

END

Media for Industrial Fermentations:

EXAMPLES OF COMMONLY USED CARBON SOURCES-1

Media for Industrial Fermentations

❑Carbohydrates

It is common practice to use carbohydrates as the carbon source in microbial fermentation processes. The most widely available carbohydrate is starch obtained from maize grain. It is also obtained from other cereals, potatoes and cassava.

Maize and other cereals may also be used directly in a partially ground state, e.g. maize chips. Starch may also be readily hydrolysed by dilute acids and enzymes to give a variety of glucose preparations (solids and syrups).

Hydrolysed cassava starch is used as a major carbon source for glutamic acid production in Japan.

Media for Industrial Fermentations

❑Carbohydrates

Syrups produced by acid hydrolysis may also contain toxic products which may make them unsuitable for particular processes.

Barley grains may be partially germinated and heat treated to give the material known as malt, which contains a variety of sugars besides starch. Malt is the main substrate for brewing beer. Malt extracts may also be prepared from malted grain.

Sucrose is obtained from sugar cane and sugar beet. It is commonly used in fermentation media in a very impure form as beet or cane molasses, which are the residues left after crystallization of sugar solutions in sugar refining.

Media for Industrial Fermentations

Carbohydrate composition of barley malt (expressed as % dry weight of total)

Starch	58-60
Sucrose	3-5
Reducing sugars	3-4
Other sugars	2
Hemicellulose	6-8
Cellulose	5

Analysis of beet and cane molasses (expressed as % of total w/v)

Sucrose	48.5	33.4
Raffinose	1.0	0
Invert sugar	1.0	21.2

Media for Industrial Fermentations

Molasses is used in the production of high-volume/low-value products such as ethanol, SCP, organic and amino acids and some microbial gums. In 1980, 300,000 tons of cane molasses were used for amino acid production in Japan (Minoda, 1986).

Molasses or sucrose also may be used for production of higher value/low-bulk products such as antibiotics, speciality enzymes, vaccines and fine chemicals.

The cost of molasses will be very competitive when compared with pure carbohydrates. However, molasses contains many impurities and molasses-based fermentations will often need a more expensive and complicated extraction/purification stage to remove the impurities and effluent treatment will be more expensive because of the unutilized waste materials which are still present.

Media for Industrial Fermentations

The use of lactose and crude lactose (milk whey powder) in media formulations is now extremely limited since the introduction of continuous-feeding processes utilizing glucose.

END

Media for Industrial Fermentations:

EXAMPLES OF COMMONLY USED CARBON SOURCES-2

Media for Industrial Fermentations

❑Oils and Fats

Oils were first used as carriers for antifoams in antibiotic processes. Vegetable oils (olive, maize, cotton seed, linseed, soya bean, etc.) may also be used as carbon substrates, particularly for their content of the fatty acids, oleic, linoleic and linolenic acid, because costs are competitive with those of carbohydrates.

Bader *et al.,* (1984) discussed factors favouring the use of oils instead of carbohydrates. A typical oil contains approximately 2.4 times the energy of glucose on a per weight basis. Oils also have a volume advantage as it would take 1.24 dm^3 of soya bean oil to add 10 kcal of energy to a fermenter, whereas it would take 5 dm^3 of glucose or sucrose assuming that they are being added as 50% w/w solutions.

Media for Industrial Fermentations

❑Oils and Fats

Ideally, in any fermentation process, the maximum working capacity of a vessel should be used. Oil based fed-batch fermentations permit this procedure to operate more successfully than those using carbohydrate feeds where a larger spare capacity must be catered for to allow for responses to a sudden reduction in the residual nutrient level Oils also have antifoam properties which may make downstream processing simpler, but normally they are not used solely for this purpose.

Media for Industrial Fermentations

Soyabean oil or **Rapeseed oil** and **Glycerol trioleate** is known to be used in some fermentations. **Methyl oleate** has been used as the sole carbon substrate in cephalosporin production.

END

Media for Industrial Fermentations:

EXAMPLES OF COMMONLY USED CARBON SOURCES-3

Media for Industrial Fermentations

❑Hydrocarbons and their derivatives

Significant research work has been done using n-alkanes for production of organic acids, amino acids, vitamins and co-factors, nucleic acids, antibiotics, enzymes and proteins. Methane, methanol and n-alkanes have all been used as substrates for biomass production.

There are some advantages and disadvantages of hydrocarbons and their derivatives as fermentation substrates, particularly with reference to cost, process aspects and purity. In processes where the feedstock costs are an appreciable fraction of the total manufacturing cost, cheap carbon sources are important.

On a weight basis n-alkanes have approximately twice the carbon and three times the energy content of the same weight of sugar.

Media for Industrial Fermentations

❑Hydrocarbons and their derivatives

Although petroleum-type products are initially impure they can be refined to obtain very pure products in bulk quantities which would reduce the amount of effluent treatment and downstream processing.

At this time the view was also held that hydrocarbons would not be subject to the same fluctuations in cost as agriculturally derived feedstocks because it would be a stable priced commodity and might be used to provide a substrate for conversion to microbial protein (SCP) for economic animal and/or human consumption.

Cost of hydrocarbons does not make them economically attractive bulk feedstocks for the production products where feedstock costs are an appreciable fraction of manufacturing costs of low-value bulk products.

Media for Industrial Fermentations

It was considered that hydrocarbons and their derivatives might have a potential role as feedstocks in the microbial production of higher value products such as intermediates, pharmaceuticals, fine chemicals and agricultural chemicals.

END

Media for Industrial Fermentations:

NITROGEN SOURCES

Media for Industrial Fermentations

❑Factors influencing the choice of N_2 source

Control mechanisms exist by which nitrate reductase, an enzyme involved in the conversion of nitrate to ammonium ion, is repressed in the presence of ammonia. For this reason ammonia or ammonium ion is the preferred nitrogen source.

In fungi that have been investigated, ammonium ion represses uptake of amino acids by general and specific amino acid permeases. In *Aspergillus nidulans*, ammonia also regulates the production of alkaline and neutral proteases. Therefore, in mixtures of nitrogen sources, individual nitrogen components may influence metabolic regulation so that there is preferential assimilation of one component until its concentration has diminished.

Media for Industrial Fermentations

❑Factors influencing the choice of N_2 source

It has been shown in many studies that antibiotic production by many micro-organisms is influenced by the type and concentration of the nitrogen source in the culture medium.

Antibiotic production may be inhibited by a rapidly utilized nitrogen source (NH_4^+, NO_3^-, certain amino acids). The antibiotic production only begins to increase in the culture broth after most of the nitrogen source has been consumed.

In shake flask media experiments, salts of weak acids (e.g. ammonium succinate) may be used to serve as a N_2 source and eradicate the source of a strong acid pH change due to chloride or sulphate ions which would be present if NH_4Cl or sulphate were used as the N_2 source.

Media for Industrial Fermentations

❑Factors influencing the choice of N_2 source

The use of complex nitrogen sources for antibiotic production has been common practice. They thought to help create physiological conditions in trophophase which favour antibiotic production in idiophase.

For example, in the production of polyene antibiotics, soybean meal has been considered a good nitrogen source because of the balance of nutrients, the low phosphorus content and slow hydrolysis.

It has been suggested that this gradual breakdown prevents the accumulation of ammonium ions and repressive amino acids. These are probably some of the reasons for the selection of ideal nitrogen sources for some secondary metabolites.

Media for Industrial Fermentations

Some of the complex nitrogenous material may not be utilized by a micro-organism and create problems in downstream processing and effluent treatment. This can be an important factor in the final choice of substrate.

END

Media for Industrial Fermentations:

EXAMPLES OF COMMONLY USED NITROGEN SOURCES

Media for Industrial Fermentations

Most industrially used micro-organisms can utilize inorganic or organic sources of nitrogen. Inorganic nitrogen may be supplied as ammonia gas, ammonium salts or nitrates. Ammonia has been used for pH control and as the major nitrogen source in a defined medium for the commercial production of human serum albumin by *Saccharomyces cerevisiae*. Ammonium salts such as ammonium sulphate will usually produce acid conditions as the ammonium ion is utilized and the free acid will be liberated.

On the other hand nitrates will normally cause an alkaline drift as they are metabolized. Ammonium nitrate will first cause an acid drift as the ammonium ion is utilized, and nitrate assimilation is repressed. When the ammonium ion has been exhausted, there is an alkaline drift as the nitrate is used as an alternative nitrogen source.

Media for Industrial Fermentations

One exception to this pattern is the metabolism of *Gibberella fujikuroi* (Borrow *et al*., 1961, 1964). In the presence of nitrate the assimilation of ammonia is inhibited at pH 2.8-3.0. Nitrate assimilation continues until the pH has increased enough to allow the ammonia assimilation mechanism to restart.

Organic nitrogen may be supplied as amino acid, protein or urea. In many instances growth will be faster with a supply of organic nitrogen, and a few microorganisms have an absolute requirement for amino acids. Amino acids are more commonly added as complex organic nitrogen sources which are non-homogeneous, cheaper and readily available. In lysine production, methionine and threonine are obtained from soybean hydrolysate since it would be too expensive to use the pure amino acids.

Media for Industrial Fermentations

Best nitrogen sources for some secondary metabolites

Product	Main nitrogen source(s)	Reference
Penicillin	Corn-steep liquor	Moyer and Coghill (1946)
Bacitracin	Peanut granules	Inskeep *et al.* (1951)
Riboflavin	Pancreatic digest of gelatine	Malzahn *et al.* (1959)
Novobiocin	Distillers' solubles	Hoeksema and Smith (1961)
Rifomycin	Pharmamedia	Sensi and Thiemann (1967)
	Soybean meal, $(NH_4)_2SO_4$	
Gibberellins	Ammonium salt and natural plant nitrogen source	Jefferys (1970)
Butirosin	Dried beef blood or haemoglobin with $(NH_4)_2SO_4$	Claridge *et al.* (1974)
Polyenes	Soybean meal	Martin and MacDaniel (1977)

Media for Industrial Fermentations

Other proteinaceous nitrogen compounds serving as sources of amino acids include corn-steep liquor, soya meal, peanut meal, cotton-seed meal.

END

Media for Industrial Fermentations:

MINERALS-1

Media for Industrial Fermentations

All micro-organisms require certain mineral elements for growth and metabolism. In many media, magnesium, phosphorus, potassium, sulphur, calcium and chlorine are essential components, and because of the concentrations required, they must be added as distinct components. Others such as cobalt, copper, iron, manganese, molybdenum and zinc are also essential but are usually present as impurities in other major ingredients.

When synthetic media are used the minor elements will have to be added deliberately. The form in which the minerals are usually supplied and the concentration ranges are given in Table (in next slide).

Media for Industrial Fermentations

TABLE *The range of typical concentrations of mineral components* (g dm^{-3})

Component	Range
*KH_2PO_4	1.0–4.0 (part may be as buffer)
$MgSO_4 \cdot 7H_2O$	0.25–3.0
KCl	0.5–12.0
$CaCO_3$	5.0–17.0
$FeSO_4 \cdot 4H_2O$	0.01–0.1
$ZnSO_4 \cdot 8H_2O$	0.1–1.0
$MnSO_4 \cdot H_2O$	0.01–0.1
$CuSO_4 \cdot 5H_2O$	0.003–0.01
$Na_2MoO_4 \cdot 2H_2O$	0.01–0.1

*Complex media derived from plant and animal materials normally contain a considerable concentration of inorganic phosphate.

Media for Industrial Fermentations

The concentration of phosphate in a medium, particularly laboratory media in shake flasks, is often much higher than that of other mineral components. Part of this phosphate is being used as a buffer to minimize pH changes when external control of the pH is not being used.

In specific processes the concentration of certain minerals may be very critical. Some secondary metabolic processes have a lower tolerance range to inorganic phosphate than vegetative growth. This phosphate should be sufficiently low as to be assimilated by the end of trophophase.

Media for Industrial Fermentations

The inorganic phosphate concentration also influences production of bacitracins, citric acid (surface culture), ergot, monomycin, novobiocin, oxytetracycline, polyenes, ristomycin, rifamycin Y, streptomycin, vancomycin and viomycin.

END

Media for Industrial Fermentations:

MINERALS-2

Media for Industrial Fermentations

Weinberg (1970) has reviewed the nine trace elements of biological interest (Atomic numbers 23-30, 42). Of these nine, the concentrations of manganese, iron and zinc are the most critical in secondary metabolism.

In every secondary metabolic system in which sufficient data has been reported, the yield of the product varies linearly with the logarithmic concentration of the 'key' metal. The linear relationship does not apply at concentrations of the metal which are either insufficient, or toxic, to cell growth. Some of the primary and secondary microbial products whose yields are affected by concentrations of trace metals greater than those required for maximum growth are given in Table (in next slide).

Media for Industrial Fermentations

TABLE *Trace elements influencing primary and secondary metabolism*

Product	Trace element(s)
Bacitracin	Mn
Protease	Mn
Gentamicin	Co
Riboflavin	Fe, Co
	Fe
Mitomycin	Fe
Monensin	Fe
Actinomycin	Fe, Zn
Candicidin	Fe, Zn
Chloramphenicol	Fe, Zn
Neomycin	Fe, Zn
Patulin	Fe, Zn
Streptomycin	Fe, Zn
Citric acid	Fe, Zn, Cu
Penicillin	Fe, Zn, Cu
Griseofulvin	Zn

Media for Industrial Fermentations

Chlorine does not appear to play a nutritional role in the metabolism of fungi. It is, however, required by some of the **halophilic bacteria**. Obviously, in those fermentations where a chlorine-containing metabolite is to be produced the synthesis will have to be directed to ensure that the non-chloro-derivative is not formed. The most important compounds are **chlorotetracycline** and **griseofulvin**.

In griseofulvin production, adequate available chloride is provided by the inclusion of at least 0.1% KCl as well as the chloride provided by the complex organic materials included as nitrogen sources.

Media for Industrial Fermentations

Some of the other chlorine containing metabolites are caldriomycin, nornidulin and mollisin.

END

Media for Industrial Fermentations:

CHELATORS

Media for Industrial Fermentations

Many media cannot be prepared or autoclaved without the formation of a visible precipitate of insoluble metal phosphates.

Gaunt *et al.* (1984) demonstrated that when the medium of Mandels and Weber (1969) was autoclaved, a white precipitate of metal ions formed, containing all the iron and most of the calcium, manganese and zinc present in the medium.

Media for Industrial Fermentations

The problem of insoluble metal phosphate(s) may be eliminated by incorporating low concentrations of chelating agents such as ethylene diamine tetraacetic acid (EDTA), citric acid, polyphosphates, etc., into the medium.

These chelating agents preferentially form complexes with the metal ions in a medium. The metal ions then may be gradually utilized by the microorganism. Gaunt *et al.* were able to show that the precipitate was eliminated from Mandel and Weber's medium by the addition EDTA at 25 mg dm^{-3}. It is important to check that a chelating agent does not cause inhibition of growth the micro-organism which is being cultured.

Media for Industrial Fermentations

In many media, particularly those commonly used in large scale processes, there may not be a need to add a chelating agent as complex ingredients such as yeast extracts or proteose peptones will complex with metal ions and ensure gradual release of them during growth.

END

Media for Industrial Fermentations

The types of antifoams mentioned in previous slides are generally added when foaming occurs during the fermentation. Because many antifoams are of low solubility they need a carrier such as lard oil, liquid paraffin or castor oil, which may be metabolized and affect the fermentation process.

END

Media for Industrial Fermentations:

GROWTH FACTORS

Media for Industrial Fermentations

Some micro-organisms cannot synthesize a full complement of cell components and therefore require preformed compounds called growth factors. The growth factors most commonly required are vitamins, but there may also be a need for specific amino acids, fatty acids or sterols. Many of the natural carbon and nitrogen sources used in media formulations contain all or some of the required growth factors. When there is a vitamin deficiency it can often be eliminated by careful blending of materials. It is important to remember that if only one vitamin is required it may be occasionally more economical to add the pure vitamin, instead of using a larger bulk of a cheaper multiple vitamin source.

Media for Industrial Fermentations

Calcium pantothenate has been used in one medium formulation for vinegar production. In processes used for the production of glutamic acid, limited concentrations of biotin must be present in the medium. Some production strains may also require thiamine.

END

Media for Industrial Fermentations:

BUFFERS

Media for Industrial Fermentations

The control of pH may be extremely important if optimal productivity is to be achieved. A compound may be added to the medium to serve specifically as a buffer, or may also be used as a nutrient source. Many media are buffered at about pH 7.0 by the incorporation of calcium carbonate (as chalk). If the pH decreases the carbonate is decomposed. Obviously, phosphates which are part of many media also play an important role in buffering. However, high phosphate concentrations are critical in the production of many secondary metabolites.

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The balanced use of the carbon and nitrogen sources will also form a basis for pH control as buffering capacity can be provided by the proteins, peptides and amino acids, such as in corn-steep liquor. The pH may also be controlled externally by addition of ammonia or sodium hydroxide and sulphuric acid.

END

Media for Industrial Fermentations:

PRECURSORS

Media for Industrial Fermentations

Some components of a fermentation medium help to regulate the production of the product rather than support the growth of the micro-organism. Such additives include precursors, inhibitors and inducers, all of which may be used to manipulate the progress of the fermentation.

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Some chemicals, when added to certain fermentations, are directly incorporated into the desired product. Probably the earliest example is that of improving penicillin yields. A range of different side chains can be incorporated into the penicillin molecule.

The significance of the different side chains was first appreciated when it was noted that the addition of corn-steep liquor increased the yield of penicillin from 20 units cm^{-3} to 100 units cm^{-3}.

Corn-steep liquor was found to contain phenylethylamine which was preferentially incorporated into the penicillin molecule to yield benzyl penicillin (Penicillin G).

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Some studies showed that addition of phenylacetic acid and its derivatives to the medium were capable of both increasing penicillin production threefold and to directing biosynthesis towards increasing the proportion of benzyl penicillin from 0 to 93% at the expense of other penicillins.

Phenylacetic acid is still the most widely used precursor in penicillin production. Some important examples of precursors are given in Table (in next slide).

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Precursors used in fermentation processes

Precursor	Product	Microorganism
Phenylacetic-acid related compounds	Penicillin G	*Penicillium chrysogenum*
Phenoxy acetic acid	Penicilin V	*Penicillium chrysogenum*
Chloride	Chlortetracycline	*Streptomyces aureofaciens*
	Griseofulvin	*Penicillium griseofulvin*
Propionate	Riboflavin	*Lactobacillus bulgaricus*
Cyanides	Vitamin B12	*Proprianobacterium, Streptomyces spp.*
β-Iononones	Carotenoids	*Phycomyces blakesleeanus*
α-Amino butyric acid	L-Isoleuchine	*Bacillus subtilis*
D-Threonine	L-Isoleuchine	*Serratia marcescens*

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Precursors used in fermentation processes (Cont..)

Precursor	Product	Microorganism
Anthranilic acid	L-Tryptophan	*Hansenula anomala*
Nucleosides and bases	Nikkomycins	*Streptomyces tendae*
Dihydronovobionic acid	Dihydronovobiocin	*Streptomyces sp.*
p-Hydroxycinnamate	Organomycin A and B	*Streptomyces organonensis*
DL-α-Amino butyric acid	Cyclosporin A	*Tolypocladium inflatum*
L-Threonine	Cyclosporin c	
Tyrosine or *p*-hydroxy-phenylglycine	Dimethylvancomycin	*Nocardia orientalis*

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Having established that the activity of penicillin lay in the side chain, and that the limiting factor was the synthesis of the side chain, it became standard practice to add side-chain precursors to the medium, in particular phenylacetic acid.

END